

8.2.1 Hydroelectric power-plants

Hydroelectric power-plants use water to produce electricity. They employ water turbines coupled with generators. In a hydroelectric power-plant, water-head is created by constructing a dam across a river or lake. From the dam, water is carried through large pipes or tunnels called penstock and allowed to fall on water turbine blades through a nozzle. The water turbine rotates due to high power water jet and thus converts the hydraulic energy into mechanical energy at the turbine shaft. The turbine drives the alternator which converts mechanical energy into electrical energy. Figure 8.2 shows the layout of a typical hydro-

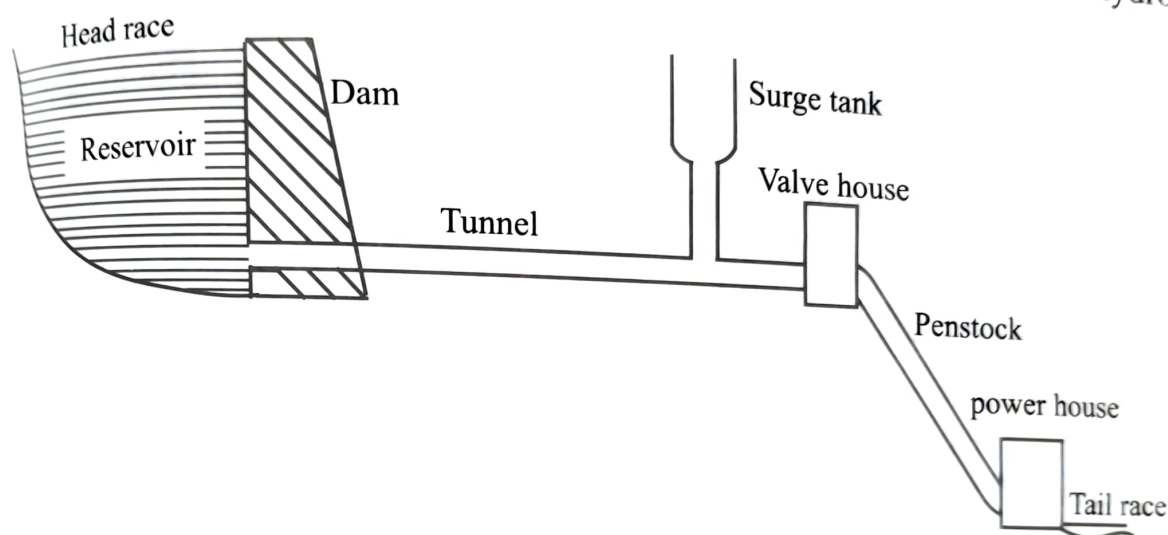


Figure 8.2: Layout of a hydroelectric power station

electric power station. The main parts are dam, tunnel, surge tank, valve house, penstock and power house.

Dam

The dam is constructed across a river or lake and forms a reservoir. It provides a steady water-head for the turbine to operate. The water level of the reservoir is called head race.

Tunnel and penstock

Tunnel and penstock form water-conduit system which carries water from the reservoir to the turbine of power house. A pressure tunnel is taken off from the reservoir, which carries water to the valve house. From the valve house, water is brought to the turbine through a huge steel pipe known as penstock. Penstocks are suitable for carrying water at medium to high pressure. For low-pressure penstocks, RCC (Reinforced Cement Concrete) pipes are also used.

Valve house

Valve house is a facility to start or stop the water flow into the power house for normal operation and maintenance. It contains main sluice valves which control the water flow to the power house and automatic isolating valves which cut off supply of water if the penstock bursts due to high pressure.

Surge tank

Surge tank is a small additional storage facility implemented just before the valve house. It is required when there is considerable distance between the power house and the reservoir. When the distance is more, non-uniform intake to the power house results in water-hammering. The surge tank allows sudden rise of water into it, when the water flow is stopped or reduced. When the water requirement of the turbine suddenly increases due to an increase in load, water from the surge tank will be taken to meet the quick demand. Thus the surge tank functions as a pressure regulator in the water-line.

Power house

Power house is normally located near the foot of the dam. Water is brought to the power house with the help of penstocks. A power house employs turbine and generator. The turbine converts the hydraulic energy from the flowing water to mechanical energy. The generator, which is mechanically coupled to the turbine, converts the mechanical energy to electrical energy. The used water from the turbine is released through the tail race. The location of the power house is determined based on the maximum possible water-head (water-head is the difference in altitude from tail race to head race). In the figure, the power house is located away from the dam and results in a higher head than at the foot of the dam.

Tailrace

Water is discharged into the tailrace after passing through the turbine, which carries into the river. A tailrace is an open channel or a tunnel depending upon the power house location. The discharge from all the turbines is collected in the tailrace by means of branch channels. The tailrace may discharge into the original river or into some other river.

Selection of site for hydroelectric power-plants

1. **Availability of water** Hydroelectric power-plants require huge amount of water. Availability of water is estimated on the basis of rainfall data and river flow data taken over a few years.

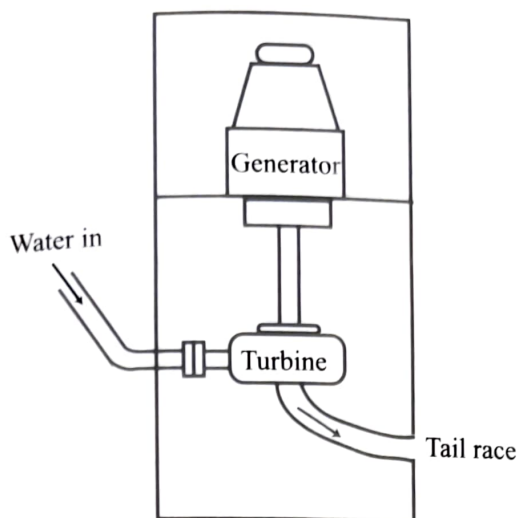


Figure 8.3: Turbine and generator in a power house

2. **Storage of water** The site should have large storage area so that water received during rainy season can be stored for the rest of the year.
3. **Head of water** The availability of head depends on the topography of the area. It should be possible to generate sufficient water-head there.
4. **Accessibility** The site should be accessible for transportation of equipments and also for power distribution.

Advantages of hydroelectric power

1. It requires no fuel, as water is used for the generation of electrical energy.
2. It is very neat and clean since no smoke or ash is produced.
3. It is comparatively simple in construction and requires less maintenance.
4. It uses simple technology.
5. It does not require a long starting time like a steam power station.
6. It is robust and has a longer life.
7. The dam and reservoir serve the purpose of irrigation and flood control etc.

Disadvantages

1. It requires large area for installation.
2. It involves high capital cost due to construction of dam.
3. There is uncertainty about the availability of huge amount of water due to dependence on weather conditions.